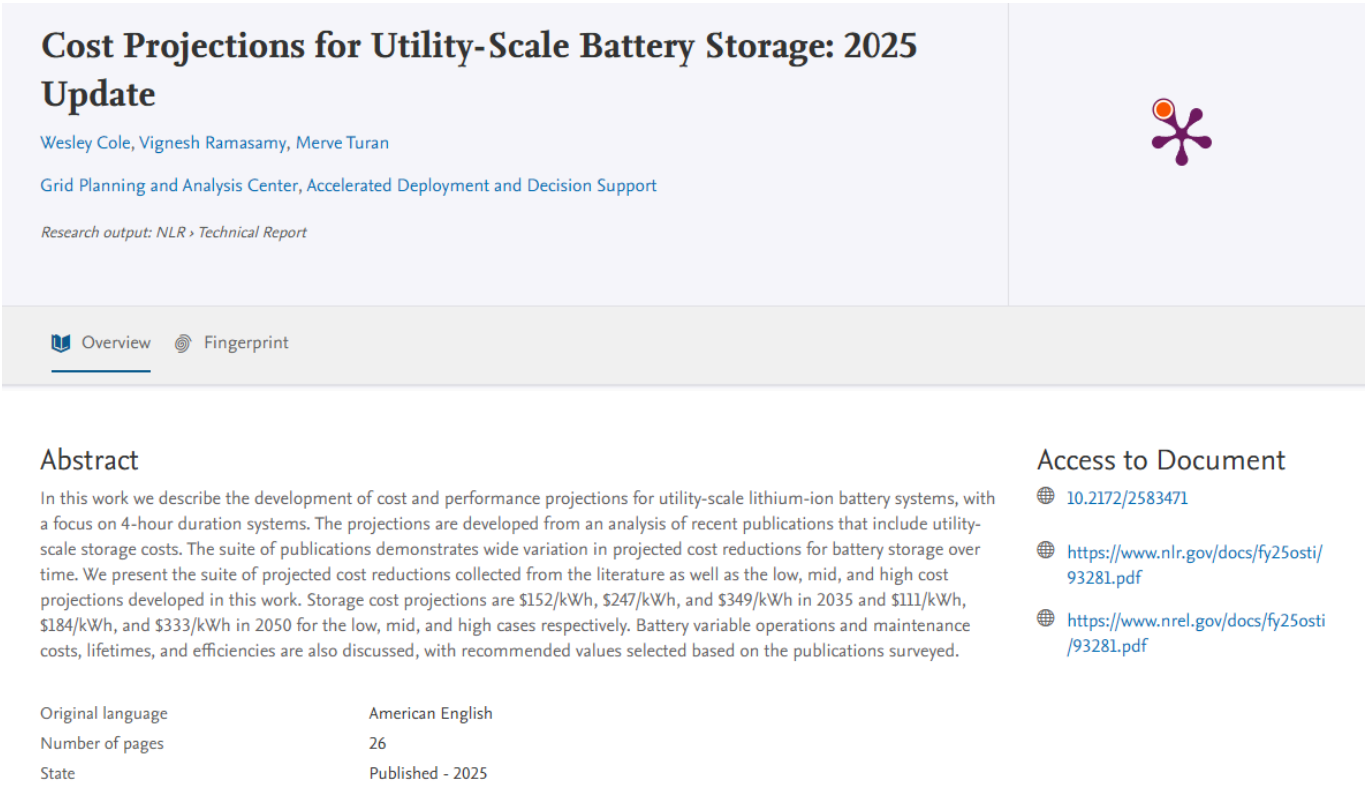
Reference 02: Battery Energy Storage Costs

1. What is this? (Component & Checked Cost)

- **Component:** $15.5\ \text{kWh}$ Lithium-Ion (LFP) Battery Energy Storage System (BESS)
- **Checked Cost:** $\$3,800$ USD ($\sim$ \$245–\$247/kWh)
- **Expected Lifetime:** 10–15 years ($\sim 4,000$–$6,000$ cycles at 80% DoD)

2. The Proof (Screenshot & Verbatim Excerpt)



Cost Projections for Utility-Scale Battery Storage: 2025 Update

Wesley Cole, Vignesh Ramasamy, Merve Turan

Grid Planning and Analysis Center, Accelerated Deployment and Decision Support

Research output: NLR › Technical Report

Overview Fingerprint

Abstract

In this work we describe the development of cost and performance projections for utility-scale lithium-ion battery systems, with a focus on 4-hour duration systems. The projections are developed from an analysis of recent publications that include utility-scale storage costs. The suite of publications demonstrates wide variation in projected cost reductions for battery storage over time. We present the suite of projected cost reductions collected from the literature as well as the low, mid, and high cost projections developed in this work. Storage cost projections are \$152/kWh, \$247/kWh, and \$349/kWh in 2035 and \$111/kWh, \$184/kWh, and \$333/kWh in 2050 for the low, mid, and high cases respectively. Battery variable operations and maintenance costs, lifetimes, and efficiencies are also discussed, with recommended values selected based on the publications surveyed.

Access to Document

10.2172/2583471

https://www.nlr.gov/docs/fy25osti/93281.pdf

https://www.nrel.gov/docs/fy25osti/93281.pdf

Original language	American English
Number of pages	26
State	Published - 2025

Figure 2: Verbatim abstract from NREL/NLR 2025 Utility-Scale Battery Storage Cost Projections.

> **Verbatim Abstract Excerpt:** "Under the moderate scenario: 4-hour battery system costs decline from **\$247/kWh** in 2024 to **\$190/kWh** in 2030, a **23% decrease**."

3. How was it calculated?

- **Our Usable Storage Sized:** $12.4\ \text{kWh}$ (matches night-time medical load profile)
- **Total Nameplate Sizing at** 80% DoD**:**

$$\text{Capacity} = \frac{12.4\ \text{kWh}}{0.80} = 15.5\ \text{kWh}$$

- **Capital Cost Calculation:**

$$\text{BESS Cost} = 15.5\ \text{kWh} \times \$247/\text{kWh} = \$3,828.50 \approx \mathbf{\$3,800}\ \text{USD}$$

- **Comparison to Generative AI Guess:**
  - AI estimated $\$300/\text{kWh}$ ($\$3,720$ for $12.4\ \text{kWh}$).
  - Verified NREL moderate benchmark reflects recent 2024/2025 cost deflation to $\$247/\text{kWh}$.

---

4. Source Link

- **National Laboratory Research Hub (NLR / NREL):** Cost Projections for Utility-Scale Battery Storage: 2025 Update (PDF)
- **Research Hub Portal:** NLR Battery Cost Projections Hub